

Project Design Phase Proposed Solution Template

Date	01 October 2026
Team ID	SWTID-2026-1665
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	- Students often struggle to understand complex concepts, create effective study plans, and assess their knowledge efficiently. - Most AI study tools need a constant internet connection, subscriptions and cloud data sharing, which many learners cannot afford or rely on.
2.	Idea / Solution description	- EduGenie is a lightweight, offline-based AI learning assistant that runs locally on the learner's device, built with a FastAPI backend and a simple HTML + CSS frontend. - It answers questions, simplifies complex concepts, generates quizzes from topics or text, summarizes educational passages, and creates structured beginner-to-advanced learning paths.
3.	Novelty / Uniqueness	- Works offline on everyday devices such as the Mac M1, so learning is not blocked by poor connectivity. - Combines Q&A, simplified explanations, quizzes, summaries and personalized learning paths in a single lightweight tool. - Study data stays on the device, protecting privacy.
4.	Social Impact / Customer Satisfaction	- Makes personalized, accessible AI-powered learning support available to students and learners at all academic levels, including areas with limited internet. - Reduces dependence on costly tutoring, builds confidence through instant feedback and quizzes, and helps learners study at their own pace.
5.	Business Model (Revenue Model)	- Freemium: free core features (Q&A, summaries, basic quizzes) for individual students. - Paid premium add-ons such as multilingual learning, voice assistance, progress tracking and adaptive quizzes. - Institutional licences for schools, colleges and coaching centres, and integration partnerships with educational platforms.
6.	Scalability of the Solution	- Lightweight local deployment means no heavy server cost per additional user. - Can be extended with new subjects, languages, voice assistance, progress tracking and adaptive quizzes. - Modular FastAPI backend allows integration with educational platforms and rollout across institutions.